

EDIDIONG DAVID IBOKETE

MACHINE LEARNING RESEARCH ENGINEER | FORECASTING | GEOSPATIAL & SCIENTIFIC ML

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Machine learning researcher and engineer with strong Python and PyTorch skills and about three years of professional experience spanning data systems, geospatial analytics, applied machine learning, and research software. Built time-series forecasting models on large weather-linked datasets, reproducible experimental pipelines, and deployed ML/AI systems with rigorous validation, failure analysis, and reusable infrastructure. Brings direct geospatial experience, first-place energy forecasting work, and co-authored statistical ML research focused on distinguishing real performance gains from noise and validating models across datasets.

TECHNICAL SKILLS

Machine Learning & Research: Python, PyTorch, Scikit-learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, regression, classification, deep learning, neural networks, time-series forecasting, anomaly detection, clustering, feature engineering, hyperparameter tuning, model evaluation, experiment design, SHAP, residual diagnostics

Data & Systems: SQL, R, SAS PROC SQL, ETL, data acquisition, cleaning, normalization, validation, multi-source integration, reproducible pipelines, Git, GitHub, REST APIs, FastAPI, CI/CD, smoke tests, configuration-driven workflows

Geospatial & Scientific Analytics: ArcGIS, geospatial data preparation and analysis, spatial clustering, weather and environmental features, scenario analysis, statistical modeling, hypothesis testing, confidence intervals, cross-dataset validation

Cloud & Engineering: Azure App Service, GitHub Actions, Azure Machine Learning, MLflow, model registration, experiment tracking, Docker, OpenTelemetry, Application Insights

EDUCATION & CERTIFICATIONS

M.S., Data Analytics (Big Data), Canisius University, Buffalo, NY | Completed December 2025

B.S., Economics, University of Abuja, Nigeria

Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300) | Azure Fundamentals (AZ-900) | Power BI Data Analyst Associate (PL-300)

PROFESSIONAL EXPERIENCE

Canisius University

Buffalo, NY

Data Analyst, Graduate Assistant

Aug 2024 - Dec 2025

- Built and maintained dependable ETL, reconciliation, and data-quality pipelines across SLATE CRM and MS SQL Server using SQL, SAS PROC SQL, R, Alteryx, and Excel, converting inconsistent multi-source records into reusable analytical datasets.
- Investigated a major cross-system record-count discrepancy by profiling source grain and business keys before deduplication, separating legitimate repeated activity from redundant records and validating each transformation stage before downstream use.
- Designed analytical data models and validation checks for recurring reporting, achieving 100% deduplication accuracy and reducing record redundancy by 70% under the agreed reporting definition.
- Partnered with admissions, enrollment, finance, and leadership stakeholders to frame ambiguous questions, test analytical assumptions, document business rules, and communicate findings through Power BI/Tableau decision-support outputs.

Center for Research Libraries

Chicago, IL

Data Scientist & Algorithm Engineer Intern

Jun 2025 - Aug 2025

- Translated an open-ended 19-institution cataloging problem into a Python/Pandas matching and allocation system spanning 194 tasks and 565 candidate matches, iterating on scoring logic as stakeholder requirements evolved.
- Found, profiled, normalized, cleaned, combined, and validated multi-source quantitative and qualitative survey data with inconsistent schemas, building dependable requester and contributor abstractions for repeated algorithm experiments.
- Designed, tested, and tuned rule-based matching logic around capacity, qualifications, reciprocity, trust, and performance, using exception analysis and transparent diagnostics to determine which changes represented genuine improvement.
- Developed modular Python and SQL components under Git version control with configuration, intermediate exports, validation checkpoints, and an end-to-end smoke test, turning experimental logic into a reproducible system for technical review and handoff.

Pear Project Inc.

Remote

Machine Learning Intern - Applied AI & Healthcare

May 2025 - Jul 2025

- Co-developed an Azure-supported conversational AI patient-intake and provider-routing prototype, translating unstructured symptom narratives into specialist recommendations against structured provider data.
- Engineered Python and R workflows to acquire, clean, transform, combine, and validate structured and unstructured healthcare inputs, including provider records, data dictionaries, and ICD-10, HCPCS, and CPT reference artifacts.
- Developed and evaluated NLP, text-classification, predictive-modeling, feature-engineering, and prompt-engineering workflows, iterating on model/output behavior through testing, quality checks, and domain-aware error analysis.
- Documented assumptions, limitations, evaluation findings, and iterative refinements for reproducible team review and handoff, bridging model development, data engineering, evaluation, and cloud-supported workflow needs.

- Prepared, integrated, analyzed, and visualized structured and geospatial datasets using Python, SQL, ArcGIS, Tableau, and Excel, surfacing spatial patterns, operational risks, anomalies, and decision-support insights.
- Administered MySQL and MS SQL Server environments with role-based access controls and data-integrity checks, building reliable data foundations for recurring analytical and geospatial workflows in mission-critical operations.
- Wrote and optimized SQL for extraction, joins, aggregation, and validation, improving query efficiency by 30%, while automating recurring calculations and reports with Excel VBA to accelerate monthly reporting by 25%.
- Joined at inception as one of two appointed technical staff, developed the centre operating manual, collaborated across technical and operational functions, and supervised more than 10 staff while maintaining data, reporting, and workflow continuity.

SELECTED MACHINE LEARNING PROJECTS

Commercial Building Energy Forecasting & Optimization | Weather-Linked Time Series | Predictive Modeling Lead | 1st Place

- Cleaned and merged hourly electricity, building metadata, and weather observations for 1,600+ commercial buildings; interpolated missing weather values, engineered lag/rolling/environmental features, and aggregated the large hourly source into a future-period daily forecasting dataset.
- Compared XGBoost with a tuned neural network using a time-based holdout; selected XGBoost at R-squared 0.9804, MAE 5.44, and RMSE 34.28, then used SHAP, residual diagnostics, and a +8 C weather stress test to probe model behavior beyond the headline score.
- Aggregated residuals by building to identify 10 underperformers and translated 10%-20% reduction scenarios into \$858K-\$1.7M in projected annual savings, connecting model diagnostics to operational decisions.

Project ARIEL | Reusable ML/AI System | FastAPI | Azure App Service

- Engineered a modular retrieval and generation system with separate ingestion, chunking, embeddings, retrieval/reranking, context assembly, evaluation, API serving, and monitoring components, then deployed the FastAPI backend to Azure App Service.
- Built a failure-driven experimentation loop using 17 retrieval suites and 277 audit questions: isolate bad retrieval, add a targeted test, patch behavior, run smoke tests, and promote successful fixes into regression coverage, achieving 100% verified top-1 and top-k retrieval accuracy.

Smart City Pattern & Anomaly Detection | Python | Spatial Clustering & Statistical Detection

- Implemented K-means from scratch for traffic-pattern clustering, compared elbow and silhouette evidence across candidate cluster counts, and selected a simpler K=3 structure over the metric-optimal K=5 when the latter produced small, less interpretable micro-clusters.
- Implemented Gaussian anomaly detection for energy-grid observations and optimized the anomaly threshold by scanning candidate probability cutoffs and selecting epsilon on validation F1, explicitly balancing detection quality against false alarms.

Customer Purchase Prediction & Conversion Analytics | Model Comparison & Threshold Tuning

- Built a leakage-safe Scikit-learn workflow across 12,205 cleaned sessions with stratified holdout evaluation, feature-type-aware preprocessing, and Logistic Regression, Random Forest, and Gradient Boosting comparisons.
- Selected Gradient Boosting at ROC-AUC 0.9377 and tuned the decision threshold to 0.70 to maximize F1 while retaining 75.13% recall, using ablation and threshold analysis to distinguish genuinely useful signals from convenient defaults.

RESEARCH

Certified Anomaly Detection for IoT Security | Co-Author, Research Software & Evaluation | 2025 - 2026

- Developed and evaluated Poisson-CI and negative-binomial concentration detectors across 359,296 primary 10-ms IoT windows, with cross-dataset validation, bootstrap/DeLong comparison, configuration-driven experiments, runtime testing, and reproducibility packaging.
- Achieved NB-CI AUC 1.0000, zero benign false positives in the strongest verified configuration, and approximately 448 microseconds per window; co-authored two submitted ML manuscripts and investigated assumptions, failure modes, and generalization across datasets.